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Foldable and/or collapsible plastic/composite utility enclosure

Abstract

A lid for a utility enclosure having a front end, a rear end, a right side, a left side, a top lid portion, and a bottom, defining an interior. Large reinforcement ribs are on an interior surface of the top lid portion and form a lattice-type structure. Small reinforcement ribs are mounted in spaces formed by the lattice type structure. The lid is made of glass fiber-reinforced polypropylene and metal bars or rebar are embedded in the large reinforcement ribs. The metal bars are positioned near a bottom of the large reinforcement ribs and spaced above the bottom. The physical structure of the lid and the lid's composition, consisting essentially of glass fiber-reinforced polypropylene, are constructed so that the lid withstands up to 33,750 pounds of force without breaking or being deformed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
533750	12/1894	Ross	N/A	N/A
941014	12/1908	Doble	N/A	N/A
1279083	12/1917	Clark	N/A	N/A
1441343	12/1922	Hartshorn	N/A	N/A
2555754	12/1950	Morin	N/A	N/A
2941710	12/1959	Smith et al.	N/A	N/A
3000049	12/1960	Terry, Jr.	N/A	N/A
3107023	12/1962	Jacobson et al.	N/A	N/A
3164281	12/1964	Williams, Jr.	N/A	N/A
3300166	12/1966	Wojciechowski	N/A	N/A
3321565	12/1966	Peterson et al.	N/A	N/A
3631584	12/1971	Walkup et al.	N/A	N/A
3878795	12/1974	Janda	N/A	N/A
3968601	12/1975	Brown et al.	N/A	N/A
D245102	12/1976	Gross	N/A	N/A
4163503	12/1978	McKinnon	N/A	N/A
4265365	12/1980	Boteler	N/A	N/A
4266670	12/1980	Mykleby	N/A	N/A
4491231	12/1984	Heggeland et al.	N/A	N/A
4579709	12/1985	Ferreri	N/A	N/A
4809402	12/1988	Rainville	N/A	N/A
4820383	12/1988	Shchamorov	220/6	B65D 11/1846
4863059	12/1988	Christensen	N/A	N/A
5288453	12/1993	Rutenbeck et al.	N/A	N/A
5333750	12/1993	McKinnon	N/A	N/A
5462383	12/1994	Van 'T Veer	N/A	N/A
6138851	12/1999	Townson	N/A	N/A

6261499	12/2000	Okeke et al.	N/A	N/A
6269966	12/2000	Pallo et al.	N/A	N/A
6325962	12/2000	Kmiecik et al.	N/A	N/A
D478672	12/2002	Charlebois	N/A	N/A
6669044	12/2002	Murakami et al.	N/A	N/A
6749080	12/2003	White	N/A	N/A
6861584	12/2004	Hutchin et al.	N/A	N/A
7030315	12/2005	Dunn et al.	N/A	N/A
7115311	12/2005	Arthurs et al.	N/A	N/A
7163352	12/2006	Jurich et al.	N/A	N/A
7234275	12/2006	Haggy et al.	N/A	N/A
7320768	12/2007	Eimura	N/A	N/A
7381888	12/2007	Burke et al.	N/A	N/A
7385137	12/2007	Burke et al.	N/A	N/A
7406240	12/2007	Murano	N/A	N/A
7501573	12/2008	Balfour, Jr.	N/A	N/A
7513385	12/2008	Flynn	N/A	N/A
7633742	12/2008	Standish et al.	N/A	N/A
7802402	12/2009	Skaggs	N/A	N/A
7807923	12/2009	Moran	N/A	N/A
7880641	12/2010	Parris et al.	N/A	N/A
7959846	12/2010	White et al.	N/A	N/A
8096439	12/2011	Fogerlie	N/A	N/A
8249411	12/2011	Burke	N/A	N/A
8261410	12/2011	Sze	N/A	N/A
8302796	12/2011	Johnson	N/A	N/A
8307604	12/2011	Stepanian et al.	N/A	N/A
8336847	12/2011	Abate	N/A	N/A
8534484	12/2012	Rost	220/237	H02G 9/10
8561641	12/2012	Ogden et al.	N/A	N/A
D693908	12/2012	Rost et al.	N/A	N/A
D693909	12/2012	Rost et al.	N/A	N/A
8686909	12/2013	Frank et al.	N/A	N/A
8796548	12/2013	Rost et al.	N/A	N/A
8847070	12/2013	Burke	N/A	N/A
9265162	12/2015	Kang	N/A	N/A
9287693	12/2015	Burke	N/A	N/A
9553433	12/2016	Lalancette	N/A	N/A
9768592	12/2016	Unger et al.	N/A	N/A
9809960	12/2016	Wiley et al.	N/A	N/A
9876340	12/2017	Unger et al.	N/A	N/A
9917427	12/2017	Rost et al.	N/A	N/A
10132052	12/2017	Dang et al.	N/A	N/A
10136530	12/2017	Dang	N/A	H05K 5/0208
D841279	12/2018	Freeman et al.	N/A	N/A
10232978	12/2018	Chinni Vergottini et al.	N/A	N/A
10256616	12/2018	Schweinberg et al.	N/A	N/A
10305268	12/2018	Unger et al.	N/A	N/A

10428487	12/2018	Cassidy et al.	N/A	N/A
10479563	12/2018	Tramm et al.	N/A	N/A
10547168	12/2019	Tipton et al.	N/A	N/A
10656363	12/2019	Elford et al.	N/A	N/A
10804644	12/2019	Cretella et al.	N/A	N/A
10840682	12/2019	Tipton et al.	N/A	N/A
11008134	12/2020	Nolen	N/A	B65D 11/1846
11008728	12/2020	Turner	N/A	N/A
11014705	12/2020	Strange	N/A	N/A
11040800	12/2020	McDade et al.	N/A	N/A
11046480	12/2020	McCrea et al.	N/A	N/A
11050229	12/2020	Baker et al.	N/A	N/A
11059666	12/2020	Lanigan et al.	N/A	N/A
11066802	12/2020	Nolen	N/A	E02D 29/00
11072904	12/2020	Chudley et al.	N/A	N/A
D941582	12/2021	Nolen et al.	N/A	N/A
11338524	12/2021	Nolen et al.	N/A	N/A
11349281	12/2021	Nolen	N/A	H02B 1/46
11374386	12/2021	Nolen	N/A	B65D 11/1846
2003/0102309	12/2002	Hartwall	N/A	N/A
2007/0158345	12/2006	Booth et al.	N/A	N/A
2008/0134466	12/2007	Massengill	N/A	N/A
2008/0190950	12/2007	Orr	220/780	B65D 43/0202
2009/0266813	12/2008	Afflerbach et al.	N/A	N/A
2010/0055389	12/2009	Allore et al.	N/A	N/A
2010/0175904	12/2009	Hebert	N/A	N/A
2010/0326001	12/2009	Herron	N/A	N/A
2012/0228302	12/2011	Byrns	N/A	N/A
2016/0297607	12/2015	Burke	N/A	C08J 5/043
2018/0100284	12/2017	Isaacson et al.	N/A	N/A
2019/0301127	12/2018	Dang et al.	N/A	N/A
2019/0345740	12/2018	Fisher et al.	N/A	N/A
2020/0018102	12/2019	Lindberg et al.	N/A	N/A
2020/0122886	12/2019	Barlier	N/A	N/A
2020/0338795	12/2019	McCulley et al.	N/A	N/A
2020/0386023	12/2019	Larson	N/A	N/A
2021/0016921	12/2020	Nolen	N/A	H02G 9/10
2021/0086979	12/2020	Guerdrum et al.	N/A	N/A
2021/0171237	12/2020	Hsu	N/A	N/A
2022/0145612	12/2021	Nolen et al.	N/A	N/A
2022/0149601	12/2021	Nolen et al.	N/A	N/A
2022/0173579	12/2021	Nolen et al.	N/A	N/A
2022/0294193	12/2021	Nolen et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	---------------------	---------	-----

2345843	12/2003	CA	N/A
2777053	12/2005	CN	N/A
102005037997	12/2006	DE	N/A
102007014415	12/2007	DE	N/A
102005016491	12/2011	DE	N/A
202020000904	12/2020	DE	N/A
1098319	12/2000	EP	N/A
1461259	12/2005	EP	N/A
2177668	12/2009	EP	N/A
2604758	12/2013	EP	N/A
1515409	12/2013	EP	N/A
2607938	12/2014	EP	N/A
2878734	12/2015	EP	N/A
3035462	12/2015	EP	N/A
2631370	12/2015	EP	N/A
2166393	12/2016	EP	N/A
2357284	12/2016	EP	N/A
2166392	12/2017	EP	N/A
3029218	12/2017	EP	N/A
3483343	12/2018	EP	N/A
3518634	12/2018	EP	N/A
3536860	12/2018	EP	N/A
3567688	12/2018	EP	N/A
3300196	12/2019	EP	N/A
2738584	12/2020	EP	N/A
2668350	12/2017	ES	N/A
2933993	12/2012	FR	N/A
2324324	12/2000	GB	N/A
2394968	12/2005	GB	N/A
2404409	12/2006	GB	N/A
2436341	12/2010	GB	N/A
2450930	12/2010	GB	N/A
2520784	12/2014	GB	N/A
2576181	12/2019	GB	N/A
2576444	12/2019	GB	N/A
20180087548	12/2017	KR	N/A
WO2014/147477	12/2013	WO	N/A
WO2015/107355	12/2014	WO	N/A
WO2016/022481	12/2015	WO	N/A
WO2016/168497	12/2015	WO	N/A
WO2020/011334	12/2019	WO	N/A
WO2020/072736	12/2019	WO	N/A
WO D210289	12/2019	WO	N/A
WO2021/042950	12/2020	WO	N/A
WO2021/107628	12/2020	WO	N/A

OTHER PUBLICATIONS

RTP Company Date Sheet found at <https://www.rtpcompany.com/technical-info/data-sheets/series-100-structural/> (Year: 2004). cited by examiner

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/731,777, filed on Apr. 28, 2022, which is a continuation of U.S. patent application Ser. No. 17/224,155, filed on Apr. 7, 2021, which is a continuation in part of U.S. patent application Ser. No. 16/663,391, filed on Oct. 25, 2019, which claims the benefit of priority to U.S. Provisional Patent Application No. 62/751,278, filed on Oct. 26, 2018, the disclosures of all of which are incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

(1) The invention relates to underground utility enclosures for providing protection for below ground devices such as water meters, valves, telecom hardware, and the like and, more particularly, to a foldable and/or collapsible plastic/composite utility enclosure.

BACKGROUND OF THE INVENTION

(2) Utility enclosures are used to protect buried meters, valves, and telecom hardware and need to be capable of withstanding heavy loads that may be placed on the cover of the utility enclosure. Concrete utility enclosures are commonly used in or near roadways and driveways since they are heavy duty and are capable of withstanding heavy loads placed on the cover. Concrete utility enclosures are formed by a wall of concrete that extends around the utility. A lip extends about the top of the utility enclosure in the interior of the utility enclosure. The cover of the utility enclosure fits into the interior of the utility enclosure and rests on the lip. The concrete lip provides sufficient support to the cover to enable the utility enclosure to withstand the force of heavy loads.

(3) Concrete utility enclosures present some disadvantages, however. Concrete utility enclosures are difficult to transport and install. They are bulky and take up space. During transport, concrete utility enclosures are prone to being chipped, and significant amounts of concrete utility enclosures become non-usable because of damage incurred in transport. Installation of concrete utility enclosures is difficult because of the weight of the utility enclosures. Installation of a concrete utility enclosure generally requires the effort of two people because the utility enclosures weigh in excess of two hundred pounds each. The weight of the concrete utility enclosures also presents some danger of physical injury to the persons installing the utility enclosure.

(4) Plastic utility enclosures are known. U.S. Pat. No. 5,333,750 discloses a plastic utility enclosure that is durable, lightweight, and capable of being located in roadways or driveways and withstanding heavy loads. However, plastic utility enclosures also are bulky and take up space during transportation. What is needed is a foldable utility enclosure that takes up reduced space when folded.

SUMMARY OF THE INVENTION

(5) This invention is a foldable plastic/composite utility enclosure having a first side, a second side opposite the first side, a third side, and a fourth side opposite the third side. Each of the sides has a first end and a second opposite end. The first end of the first side is connected to the first end of the third side by means of a first hinge. The second opposite end of the second side is connected to the second opposite end of the third side by means of a second hinge. The second opposite end of the first side is connected to the second opposite end of the fourth side by means of a third hinge. The

first end of the second side is connected to the first end of the fourth side by means of a fourth hinge.

(6) The third side is constructed to rotate on the first and second hinges only away from the first side. The fourth side is constructed to rotate on the third and fourth hinges only towards the first side. The third side and the fourth side are constructed to rotate until the first side engages the second side and the foldable plastic/composite utility enclosure is in a folded configuration. The folded plastic/composite utility enclosure when in a folded configuration has only 15 to 20 percent of the thickness compared to an unfolded configuration.

(7) The first end of the first side and the first end of the second side are curved. The first end of the third side is positioned slideably over the first end of the first side and the first end of the fourth side is positioned slideably over the first end of the second side. The second opposite end of the first side abuts against the second opposite end of the fourth side and the second opposite end of the third side abuts against the second opposite end of the second side. The second opposite ends are angled.

(8) Each hinge consists of hinge members wherein each hinge member is positioned in a housing. The housing has an open interior to help anchor the housing in the ground. Each edge of each side has a plurality of spaced-apart hinge members wherein hinge members on one side engage the hinge members on another side to form the hinge. One or more of the hinges may have a reversibly insertable locking pin that prevents the sides from rotating on the hinges and prevents the foldable plastic/composite utility enclosure from folding.

(9) The foldable and/or collapsible plastic/composite utility enclosure of this invention can be made of any type of plastic or composite, preferably plastic imbedded with fibers such as carbon fibers, glass fibers, or ceramic fibers or imbedded with particles such as glass particles. The use of plastic/composite and various openings or cutouts makes the utility enclosure light weight. The hinges allow the utility enclosure to be folded flat for storage and transportation. The pairs of sides are symmetrical which facilitates manufacturing. The foldable plastic/composite utility enclosure is ideal for housing underground utilities. However, it can be used to house or contain any desired device.

(10) A preferred embodiment of a lid for the utility enclosure has a front end, a rear end, a right side, a left side, a top lid portion, and a bottom, defining an interior. First reinforcement ribs are on an interior surface of the top lid portion. Metal bars are embedded in the first reinforcement ribs. The lid is made, preferably, of glass fiber-reinforced polypropylene.

(11) The first reinforcement ribs form a lattice-type structure, with second reinforcement ribs mounted in spaces formed by the lattice type structure. The metal bars are positioned in the first reinforcement ribs near a bottom of the first reinforcement ribs and are spaced above the bottom of the first reinforcement ribs. The top lid portion has a recess with a lift pin therein. A lifting tool can be inserted into the recess and under the lift pin to lift the lid off a utility enclosure. The top lid portion may have bolt holes to bolt the lid to the utility enclosure. The lid is constructed to withstand up to 33,750 pounds of force without breaking or being deformed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top, front, perspective view of the utility enclosure of the present invention.
- (2) FIG. 2A is perspective view from an exterior surface of a first side the utility enclosure.
- (3) FIG. 2B is a perspective view from an interior surface of the first side.
- (4) FIG. 3A is perspective view from an exterior surface of a third side.
- (5) FIG. 3B is a perspective view from an interior surface of the third side.
- (6) FIG. 4 is a top, interior surface view of the second side of the utility enclosure.

- (7) FIG. 5 is a top, interior surface view of the third side of the utility enclosure.
- (8) FIG. 6 is a top, exterior surface view of the first side of the utility enclosure.
- (9) FIG. 7A is a top perspective view of the utility enclosure laying with its first side on a horizontal plane in an unfolded configuration.
- (10) FIG. 7B shows the utility enclosure of FIG. 7A in a semi-folded configuration.
- (11) FIG. 7C shows the utility enclosure of FIG. 7A in a completely folded configuration.
- (12) FIG. 8A shows an enlarged top view of an end of a second side and an end of the third side in an unfolded configuration.
- (13) FIG. 8B show an enlarged top view of the end of the second side and an end of a fourth side in an unfolded configuration.
- (14) FIGS. 9A and 9B show the same views as 8A and 8B but with the utility enclosure in a semi-folded configuration.
- (15) FIGS. 10A and 10B show the same views 8A and 8B but with the utility enclosure in a completely folded configuration.
- (16) FIG. 11 shows a top perspective view of the utility enclosure with a top cover inserted in a top end of the utility enclosure.
- (17) FIG. 12 shows a top perspective view of the top cover having an opening with a door.
- (18) FIG. 13 shows a bottom perspective view of the utility enclosure with a bottom cover attached reversibly to a bottom end of the utility enclosure.
- (19) FIG. 14 shows a perspective view of corner with a hinge connecting two sides, with a locking pin inserted through the hinge members to prevent rotation of the sides on the hinge and to prevent folding of the foldable plastic utility enclosure.
- (20) FIG. 15 shows a top view of a preferred embodiment of a lid for the utility enclosure of the present invention.
- (21) FIG. 16 shows a bottom view of the lid.
- (22) FIG. 17 shows a rear, bottom, perspective view of the lid.
- (23) FIG. 18 shows a front-end view of the lid.
- (24) FIG. 19 shows a rear, top, perspective, sectional view of the lid.
- (25) FIG. 20 shows a bottom perspective sectional view of the lid.
- (26) FIG. 21 shows an enlarged view of rebar positioned in a large reinforcement rib of the lid.
- (27) FIG. 22 illustrates a testing device for measuring the load strength of the lid as described in ANSI/SCTE 77.

DETAILED DESCRIPTION OF THE INVENTION

- (28) While the following description details the preferred embodiments of the present invention, it is to be understood that the invention is not limited in its application to the details of arrangement of the parts or the illustrations disclosed herein, since the invention is capable of other embodiments and of being practiced in various ways.
- (29) FIG. 1 is a top, front, perspective view of the utility enclosure 10 of the present invention. The utility enclosure 10 is rectangular and has a first side 11, a second side 12 opposite the first side 11, a third side 13, and a fourth side 14 opposite the third side 13. The first side 11 and the second side 12 form a first pair of sides and the third side 13 and the second side 14 form a second pair of sides. The utility enclosure 10 has a top end 15, a bottom end 16, and an interior 17. An interior ledge 18 is formed on each side in the interior 17 near the top end 15 to support a cover (see FIGS. 11 and 12) or an extension enclosure 80 (See FIG. 16). The edges of each side 11, 12, 13, and 14 are joined together with hinges 20, 21, 22, and 23 to form the rectangular utility enclosure 10, with a hinge in each corner of the rectangle. The hinges 20, 21, 22, and 23 are made up of hinge members 24. Each hinge member 24 has a central hole 25 (see FIGS. 2A and 2B) for the insertion of a hinge pin 26 to form the hinge, as is known in the art. The hinge members 24 are formed in open hollow housings 27, the housings 27 having interiors 28.
- (30) First side 11 has a first end 30 and a second opposite end 31. The second side 12 has a first end

32 and a second opposite end 33. The third side 13 has a first end 34 and a second opposite end 35. The fourth side 14 has a first end 36 and a second opposite end 37. The first ends 30 and 32 of the sides 11 and 12 (first pair of sides) form curved corners which end with curved extensions 38. The second opposite ends 31 and 33 of sides 11 and 12 are also curved and form acute angles 39. The first ends 34 and 36 of sides 13 and 14 (second pair of sides) and the second opposite ends 35 and 37 of sides 13 and 14 form obtuse angles 40. The first ends 34 and 36 of the sides 13 and 14 fit over the curved extensions 38 of sides 11 and 12 and the first ends 34 and 36 may be slightly curved to match the curved extensions 38. The second opposite end 35 of the third side 13 forms an obtuse angle 40 between an interior surface 45 (see FIG. 3B) of the third side 13 and an interior surface 42 (see FIG. 2B) of the first side 11. The obtuse angle 40 extends from the top end 15 of the utility enclosure 10 to the bottom end 16 of the utility enclosure 10. The second opposite end 37 of the fourth side 14 forms an acute angle 39 between an interior surface 45A of the fourth side 14 and an interior surface 42 of the first side 11 and the acute angle 39 extends from the top end 15 of the utility enclosure 10 to the bottom end 16 of the utility enclosure 10. The angled edges of the second opposite ends 35 and 37 of sides 13 and 14 abut the angled edges of the ends 31 and 33 of the sides 11 and 12. The sides 11, 12, 13, and 14 may have openings 19 to accommodate wires, pipes, tubing, and the like.

(31) FIG. 2A is perspective view from an exterior surface 41 of the first side 11, separated from the utility enclosure 10, and FIG. 2B is a perspective view from an interior surface 42 of the first side 11. The hinge members 24, the housings 27, and the interiors 28 can be seen in more detail. Grooves or spaces 43 between the hinge members 24 accommodate the housings 27 of the hinge members 24 that engage the first side 11 to complete the formation of the hinges 23.

(32) FIG. 3A is perspective view from an exterior surface 44 of the third side 13 separated from the utility enclosure 10 and FIG. 3B is a perspective view from an interior surface 45. The hinge members 24, the housings 27, and the interiors 28 can be seen in more detail. FIGS. 3A and 3B further show latches 46 which lock the third side 13 to the first side 11 to prevent the utility enclosure 10 from folding. Grooves 47 between the hinge members 24 accommodate the housings 27 of the hinge members 24 that engage the third side 13 to complete the formation of the hinges 20, 21, 22, 23.

(33) The housings 27 strengthen the hinges 20, 21, 22, 23 and protect them from external forces. When the utility enclosure 10 is placed in the ground to protect an underground device, soil, rocks, and cement can fill the interiors 28 of the housings 27 which will prevent the utility enclosure 10 from moving or folding. The first side 11 and the second side 12 are identical to each other and the third side 13 and the fourth side 14 are identical to each other with regard to hinging and folding features.

(34) FIG. 4 is a top, interior surface view of the second side 12. FIG. 5 is a top, interior surface view of the third side 13. FIG. 6 is a top, exterior surface view of the first side 11. The utility enclosure 10 is formed by inserting the housings 27 on end 35 of the third side 13 in between the housings 27 on end 31 of the second side 12 and inserting the housings 27 on end 34 of the third side 13 in between the housings 27 on end 30 of the first side 11. The pins 26 are then inserted through the holes 25 to form the hinges 20, 21, 22, 23. The fourth side 14 is added to sides 11 and 12 in the same way to complete the assembly of the utility enclosure 10.

(35) FIG. 7A is a top perspective view of the utility enclosure 10 laying with its first side 11 on a horizontal plane, shown by double arrow 50. The utility enclosure 10 is in an unfolded configuration. FIG. 7B shows the utility enclosure 10 in a semi-folded configuration. FIG. 7C shows the utility enclosure 10 in a completely folded configuration. As the second side 13 is rotated away from the first side 11 the fourth side 14 is rotated towards the first side 11. The second side 12 moves inward towards the first side 11. In the folded configuration the utility enclosure 10 has about 15% to 25% of the thickness, compared to the unfolded configuration, preferably about 20%.

(36) FIG. 8A shows an enlarged top view of the end 33 of the second side 12 and the end 35 of the

third side **13** in an unfolded configuration. FIG. **8B** shows an enlarged top view of the end **32** of the second side **12** and the end **36** of the fourth side **14** in an unfolded position. The utility enclosure **10** cannot fold in a direction opposite to that shown in FIGS. **9A**, **9B**, **10A**, and **10C** because the end **35** of side **13** abuts the end **33** of side **12**, shown by arrow **51**, at an angle relative to sides **12** and **13**. The end **35** of side **13** cannot move around the end **33** of side **12**. However, the end **36** of side **14** can move around end **32** of side **12**, as shown in FIGS. **9A**, **9B**, **10A**, and **10C**, because end **36** overlaps end **32**, shown by arrow **52**.

(37) FIGS. **9A** and **9B** show the same views as FIGS. **8A** and **8B** but with the utility enclosure **10** in a semi-folded configuration. Arrow **53** in FIG. **9A** shows the extent to which end **35** of side **13** has moved away from end **33** of side **12**. Arrow **54** in FIG. **9B** shows the extent to which end **36** of side **14** has moved over and around the curved extension **38** of end **32** of side **12**. FIGS. **10A** and **10B** show the same views as FIGS. **8A** and **8B** but with the utility enclosure **10** in a completely folded configuration with the first side **11** engaging the second side **12**. The folding of the utility enclosure **10** is reversible and the utility enclosure **10** can be restored to its unfolded configuration by pulling the first side **11** and the second side **12** apart.

(38) FIG. **11** shows a top perspective view of the utility enclosure **10** with a top cover **60** inserted in the top end **15**. The top cover **60** can have handles or openings for removing the cover **60**. FIG. **12** shows a top perspective view of a top cover **61** having an opening **62** with a door **63**. The interior **17** of the utility enclosure **10** can be viewed and accessed by raising the door **63**.

(39) FIG. **13** shows a bottom perspective view of the utility enclosure **10** with a bottom cover **64** attached reversibly to the bottom end **16** by methods known in the art. The use of a bottom cover **64** allows the utility enclosure **10** to be used as any type of utility enclosure for any type of articles.

(40) FIG. **14** shows a perspective view of corner with a hinge connecting two sides, with a locking pin **70** inserted through the hinge members **24** to prevent rotation of the sides on the hinge and to prevent folding of the foldable plastic/composite utility enclosure. An insertion hole **71** is on the top end **15** to insert the locking pin **70**. Each hinge member **24** has an upper groove **72** and a bottom groove **73** through which the locking pin **70** passes. The locking pin **70** can be retained in an opening **74** in the bottom hinge member.

(41) Top covers or lids for utility enclosures, under certain circumstances, may require sufficient reinforcement so that they may withstand several thousand pounds of pressure. To achieve sufficient reinforcement the lid may be made with strong plastic material, have a suitable thickness, and contain rebar. FIG. **15** shows a top view of a preferred embodiment of a lid **100** for the utility enclosure of the present invention. The lid **100** has a front end **101**, and a rear end **102**, a right side **103**, a left side **104**, and a top lid portion **105**, defining an interior **113** (see FIG. **16**). There are two recesses **106** on the top lid portion **105**. Within the recess **106** is a lift pin **107**. A lifting tool can be inserted into the recess **106** and under the lift pin **107** to lift the lid **100** off a utility enclosure. Each side **103** and **104** has bolt holes **108** for bolting the lid **100** to the top of a utility enclosure.

(42) FIG. **16** shows a bottom view of the lid **100**. The lid **100** has an open bottom **109** with large size reinforcement ribs **110** attached to an interior surface **116** of the top lid portion **105**. The large size reinforcement ribs **110** extend from the front end **101** to the rear end **102** and from the right side **103** to the left side **104**. The large size reinforcement ribs **110** crisscross forming a lattice-type structure, with medium size reinforcement ribs **111** and small size reinforcement ribs **112** mounted in spaces formed by the lattice type structure. The large size reinforcement ribs **110** are larger than the medium size reinforcement ribs **111** and the small size reinforcement ribs **112**, and the medium size reinforcement ribs **111** are larger than the small size reinforcement ribs **112**.

(43) FIG. **17** shows a rear, bottom, perspective view of the lid **100**. FIG. **17** further illustrates the interior surface **116** and the lattice-type structure, with medium size and small size reinforcement ribs in the spaces of the lattice-type structure.

(44) FIG. **18** shows a front-end view of the lid **100**. FIG. **18** shows that the bottom **114** of the large reinforcement ribs **110** extends below the bottom **109** of the lid **100**. The large reinforcement ribs

110 must be of sufficient size to sufficiently reinforce the lid to resist 33,750 pounds of force.

(45) FIG. **19** shows a rear, top, perspective, sectional view of the lid **100**. The recess **106** is of sufficient size so that a lift tool can be inserted into the recess **106**. Lift pin **107** extends across the recess **106** so that a lift tool can be positioned under the lift pin **107** and can lift the lid **100** off the top of a utility enclosure.

(46) FIG. **20** shows a bottom, perspective, sectional view of the lid **100**. FIG. **20** further shows the positioning of metal rods or rebar **115** in the large reinforcement ribs **110**. The rebar **115** is near the bottom end **114** of the large reinforcement ribs **110** but is positioned above the bottom end **114**. During the molding process of the lid **100** the rebar tends to settle to the bottom **114** of non-solidified large reinforcement ribs **110**. That settling of the rebar **115** would prevent the rebar **115** from being properly encased in the large reinforcement ribs **110**. To prevent this settling a stud cradle is used in a mold to support the rebar **115** above the bottom **114** of the large reinforcements rib **110**. When the lid **100** is removed from the mold after curing the stud cradle remains in the mold.

(47) FIG. **21** shows an enlarged view of rebar **115** positioned in a large reinforcement rib **110** of the lid **100** near the bottom **114** of the large reinforcement rib **110** and spaced above the bottom **114**. FIG. **21** further shows a void **117** of a stud cradle which is removed from the lid **100** when the lid **100** is removed from the mold.

(48) The lid **100** is made with a thermoplastic, preferably with polypropylene containing glass fiber and preferably in the amount of 15% to 40%. Glass fiber-reinforced polypropylene has improved dimensional stability, resistance to warpage, rigidity, and strength. Heat deflection temperature at 264 psi is increased up to 300° F. (150° C.) for 40% glass fiber reinforced polypropylene. Polypropylene's coefficient of thermal expansion is cut in half with 40% glass reinforcement. Glass fiber-reinforced polypropylene has good impact strength, very good mold filling capability, excellent stiffness, good dimensional stability, and low creep under load at elevated temperatures (RTP Company, Winona, Minnesota). The physical structure of the lid as described above and its composition consisting essentially of glass fiber-reinforced polypropylene are constructed so that the lid withstands up to 33,750 pounds of force without breaking or being deformed.

(49) FIG. **22** illustrates a testing device for measuring the load strength of the lid as described in ANSI/SCTE 77. The lid **100** is tested while sitting on the interior ledge **18** of the utility enclosure **10**. A rubber load plate **120** is centered on the top **105** of the lid **100** and a 10-inch by 10-inch by 1-inch steel load plate (Tier **15**) or 10-inch by 20-inch by 1-inch steel load plate (Tier **22**) **121** is centered on the rubber plate **120**. A load frame **124** is centered over the top portion **105** and a hydraulic load device **123** engages the 1-inch steel load plate **121**. Increasing force, perpendicular to the top lid portion **105**, is applied to a center of the steel plate **121** until the lid **100** bends or is deformed. Linear measurement gauges **125** and **126** are mounted to a measurement gauge mounting arm **127** to measure the amount of displacement to the top **105** of lid **100**.

(50) The foregoing description illustrates and describes the disclosure. Additionally, the disclosure shows and describes only the preferred embodiments but, it is to be understood that the preferred embodiments are capable of being formed in various other combinations, modifications, and environments and are capable of changes or modifications within the scope of the invention concepts as expressed herein, commensurate with the above teachings and/or the skill or knowledge of the relevant art. The embodiments described herein above are further intended to explain the best modes known by applicant and to enable others skilled in the art to utilize the disclosure in such, or other, embodiments and with the various modifications required by the particular applications or uses thereof. Accordingly, the description is not intended to limit the invention to the form disclosed herein. Also, it is intended that the appended claims be construed to include alternative embodiments. It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated above in order

to explain the nature of this invention may be made by those skilled in the art without departing from the principle and scope of the invention as recited in the following claims.

Claims

1. An enclosure assembly, comprising: a foldable enclosure, the foldable enclosure comprising a plurality of sides, each of the plurality of sides having a first end and a second opposite end, wherein the first end of each of the plurality of sides is connected to the second opposite end of another of the plurality of sides, and wherein the foldable enclosure is movable between an unfolded position and a folded position; and a removable lid formed from a thermoplastic matrix, the removable lid comprising a plurality of first reinforcement ribs extending between a front end and rear end and between a first side and a second side and forming a lattice-type structure, wherein a plurality of second reinforcement ribs are mounted in spaces formed by the lattice-type structure, wherein the plurality of first reinforcement ribs are larger than the plurality of second reinforcement ribs such that the plurality of first reinforcement ribs extend to a greater depth from a bottom surface of the lid than the plurality of second reinforcement ribs, wherein the lid is constructed to withstand up to 33,750 pounds of force without breaking or being deformed.
2. The enclosure assembly of claim 1, wherein the plurality of sides comprises a first side, a second side opposite the first side, a third side, and a fourth side opposite the third side, the first end of the first side connected to the first end of the third side; the second opposite end of the second side connected to the second opposite end of the third side; the second opposite end of the first side connected to the second opposite end of the fourth side; and the first end of the second side connected to the first end of the fourth side.
3. The enclosure assembly of claim 2, wherein the first end of the third side at a top of the foldable enclosure overlaps the first end of the first side at the top of the foldable enclosure and the first end of the fourth side at the top of the foldable enclosure overlaps the first end of the second side at the top of the foldable enclosure when the foldable enclosure is in a folded position.
4. The enclosure assembly of claim 2, wherein the third side and the fourth side are rotatable until the first side engages the second side in the folded position.
5. The enclosure assembly of claim 2, wherein the third side rotates only away from the first side and the fourth side rotates only towards the first side when the foldable enclosure is moved from an unfolded position to the folded position.
6. The enclosure assembly of claim 1, wherein in the folded configuration the foldable enclosure has about 15% to 25% of the thickness than the foldable enclosure in the unfolded configuration.
7. The enclosure assembly of claim 1, wherein the first end of each of the plurality of sides is connected to the second opposite end of another of the plurality of sides by a hinge.
8. The enclosure assembly of claim 7, wherein at least one hinge includes a hinge pin.
9. The enclosure assembly of claim 7, wherein at least one hinge includes a locking pin.
10. The enclosure assembly of claim 1, wherein the foldable enclosure is formed from a composite material.
11. The enclosure assembly of claim 1, wherein the removable lid is formed from a glass-fiber reinforced thermoplastic.
12. The enclosure assembly of claim 1, wherein the plurality of first and second reinforcement ribs are provided on an interior side of the removable lid.
13. The enclosure assembly of claim 1, wherein the removable lid further includes metal reinforcements positioned in the plurality of first reinforcement ribs.
14. The enclosure assembly of claim 13, wherein the metal reinforcements are embedded in the plurality of first reinforcement ribs and positioned above a bottom end of the plurality of first reinforcement ribs extending to the greater depth from the bottom surface of the lid than the plurality of second reinforcement ribs.

15. The enclosure assembly of claim 1, wherein the removable lid comprises a recess with a lift pin therein.
